ULTRASOUND IMAGING

What is ultrasound imaging?

- **Ultrasound imaging** uses high-frequency sound waves to create real-time images of structures inside the body.
- A handheld device called a **transducer** sends sound waves into the body and detects the echoes that bounce back from tissues.
- Different tissues reflect sound differently, allowing muscles, tendons, fat, and bone to be distinguished in the image.
- **B-mode (brightness mode) ultrasound** is the most common form of ultrasound imaging. The brightness of each pixel represents the strength of the reflected sound wave.
- B-mode ultrasound produces a two-dimensional cross-sectional image of the tissues beneath the skin.
- Unlike X-rays, ultrasound does **not use ionising radiation**, making it a safe and widely used imaging technique.
- Because images are generated in real time, ultrasound allows us to observe muscles as they move and contract.

What can you see?

- Individual muscles can be identified by their characteristic shape and internal structure.
- The thin layers of connective tissue within and between muscles appear as bright lines in the image.
- Tendons and bones can also be visualised, although bone reflects most of the ultrasound signal and appears as a bright boundary.
- Changes in muscle shape can be observed as muscles contract and relax.
- Ultrasound can reveal how muscles move beneath the skin, even when little or no movement is visible externally.

Why is ultrasound useful?

- Ultrasound is widely used in **sports science**, **biomechanics**, **rehabilitation**, and **clinical medicine**.
- Researchers use ultrasound to investigate muscle architecture, tendon behaviour, and movement mechanics.
- Clinicians use ultrasound to assess muscle and tendon injuries and monitor recovery.
- Ultrasound helps us better understand how muscles produce movement and adapt to training, ageing, and disease.

How to set up this demo:

1. Plug the power cable to the into the back of the ultrasound machine.

2. Plug the linear transducer (flat not curved) into the side of the ultrasound system and lock it with the switch on the ride side of it.

3. Turn on the ultrasound system by pressing the power button (top right of key panel). Wait until the system has booted.
4. Select the "DEFAULT" mode with the mouse wheel and using the grey button marked with a star to click.
5. Use the "Freeze" button on the bottom right to go to live imaging mode or to pause the live imaging. This will freeze the last image on the screen.
6. You can split the screen into two or four compartments using the three buttons on the bottom right.


Voluson·i
2D Main
User Program
Vein
Artery
Art. sens.
Vessel Near
Vertebralis
Vessel THI
Vessel CRI
DEFAULT
XBeam-CRI
SRI Low
SRI High
Trapezoid
XTD
Sub Menu
Change
Run
Freeze
Exit
Patient
Archive
Report
Auto
Clear
CFM
PW
2D
4D
P1
P2
P3